



Build Complex Data Transformations with Code Extensions

Alba Rivas
Principal Developer
Advocate,
Salesforce

[/in/alba-rivas](#)

Rohit Dar
Senior Director
Product Management,
Salesforce

[/in/rohitdar](#)



Forward-looking statements

This presentation contains forward-looking statements about, among other things, trend analyses and statements regarding future events, anticipated growth and industry prospects, and our strategies, expectation or plans regarding product releases and enhancements. The achievement or success of the matters covered by such forward-looking statements involves risks, uncertainties and assumptions. If any such risks or uncertainties materialize or if any of the assumptions prove incorrect, results or outcomes could differ materially from those expressed or implied by these forward-looking statements. The risks and uncertainties referred to above include those factors discussed in Salesforce's reports filed from time to time with the Securities and Exchange Commission, including, but not limited to our ability to meet the expectations of our customers; uncertainties regarding AI technologies and their integration into our product offerings; the effect of evolving domestic and foreign government regulations; regulatory developments and regulatory investigations involving us or affecting our industry; our ability to successfully introduce new services and product features, including related to AI and Agentforce; our ability to execute our business plans; the pace of change and innovation and our ability to compete in the markets in which we participate; and our ability to maintain and enhance our brands.

Coffee on us.

The first 4,000 attendees to provide feedback on this event will receive a \$5 Starbucks gift card.*

1. Download the Salesforce Events mobile app.
2. Navigate to surveys.
3. Complete (4) session surveys and the Event Survey.
4. Redeem your gift at Badge Pickup in-person on the last day of the event.

*Restrictions apply. See terms and conditions at sforce.co/survey-terms





Thank you

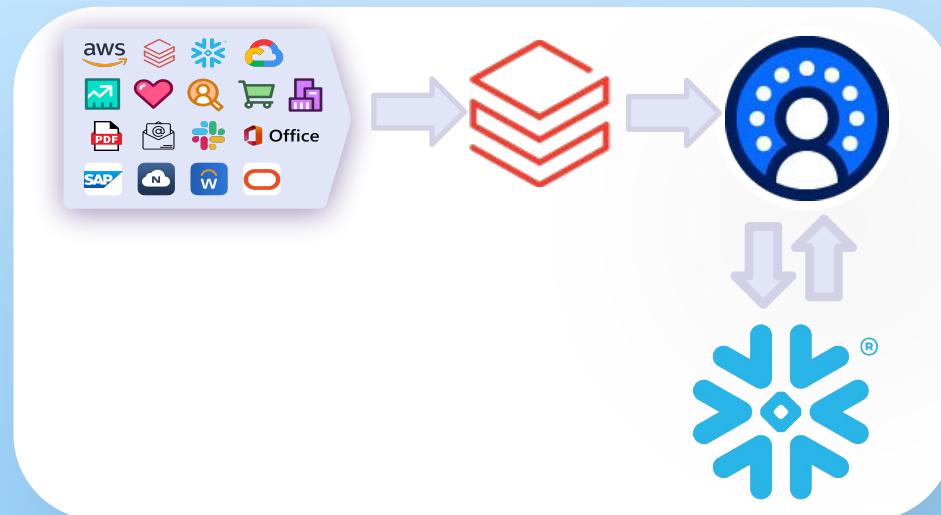


Pro-code Customization in Data 360

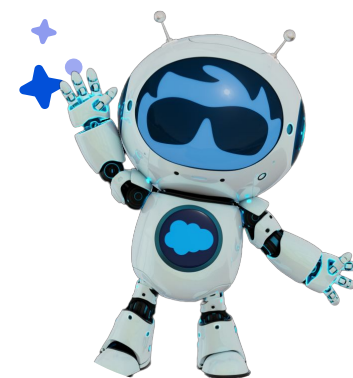
salesforce

Pro-code customizations simplify pipelines and provide improved Agentic response

Advanced data processing
performed externally



Agents need Trusted
Enterprise Context



Pro Dev users are asking for
control and customizations



The Solution: Data 360 Code Extension

salesforce

Extend Data 360 with the Power of Code

Pro-code Isolated Compute

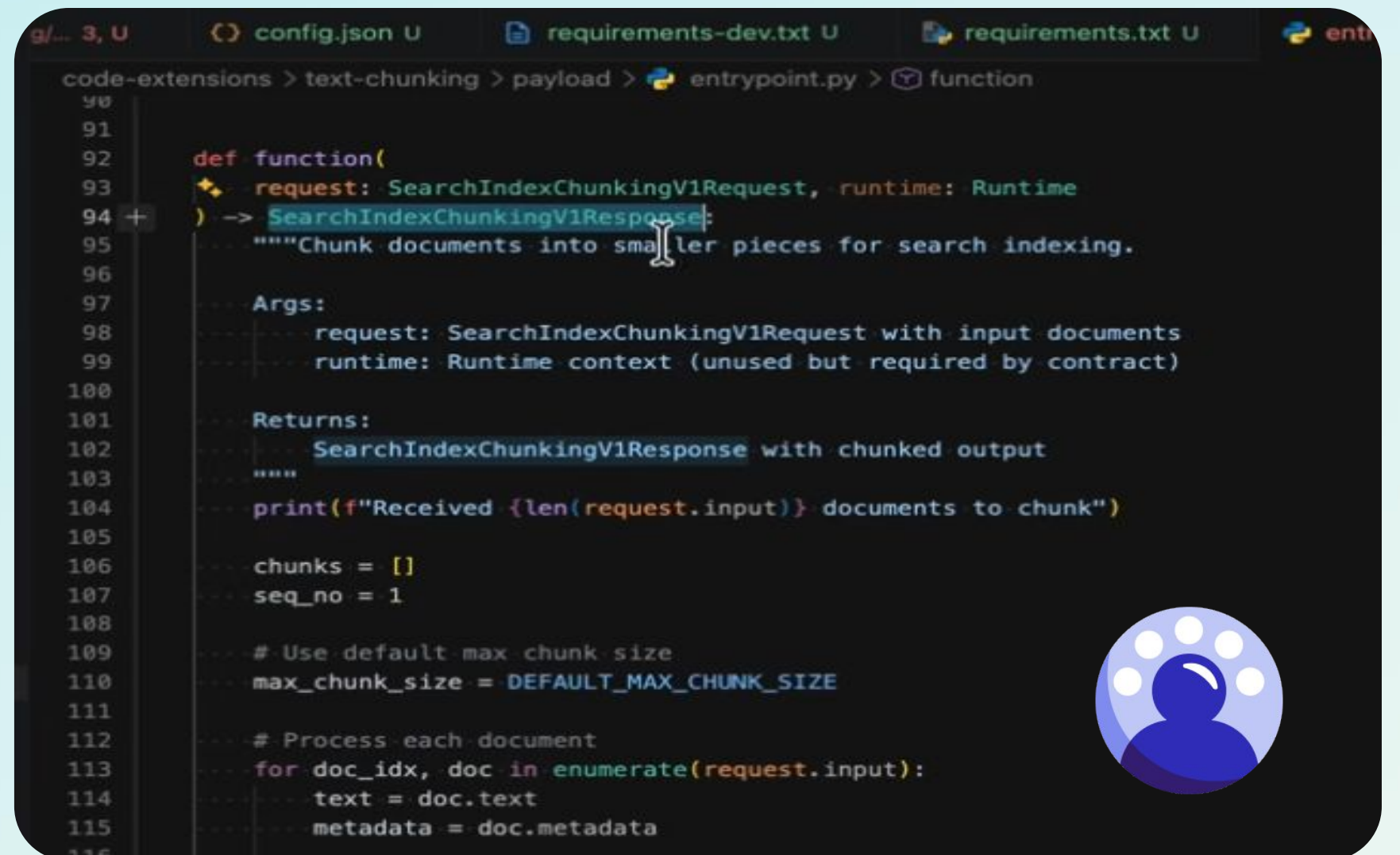
Enable developers to customize Data 360 using Python and their preferred dev tools

Data Transforms and Chunking

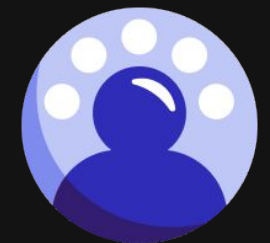
Implement Python scripts for custom data transforms and functions for chunking algorithms

Securely Execute Model and API Calls

Invoke LLMs, AI/ML models and registered API endpoints from Code Extensions



```
code-extensions > text-chunking > payload > entri
90
91
92 def function(
93     request: SearchIndexChunkingV1Request, runtime: Runtime
94 ) -> SearchIndexChunkingV1Response:
95     """Chunk documents into smaller pieces for search indexing.
96
97     Args:
98         request: SearchIndexChunkingV1Request with input documents
99         runtime: Runtime context (unused but required by contract)
100
101     Returns:
102         SearchIndexChunkingV1Response with chunked output
103     """
104     print(f"Received {len(request.input)} documents to chunk")
105
106     chunks = []
107     seq_no = 1
108
109     # Use default max chunk size
110     max_chunk_size = DEFAULT_MAX_CHUNK_SIZE
111
112     # Process each document
113     for doc_idx, doc in enumerate(request.input):
114         text = doc.text
115         metadata = doc.metadata
116
```



Code Extension Script -
Transform

GA | May 2026

Code Extension Function
- Chunking

GA | Jun 2026

LLM Gateway / AI Model

GA | Jul 2026

External API Endpoints

GA | Oct 2026

Code Extension Script -
Streaming Transform

GA | Oct 2026

Customers Can Extend Data 360 With Code Using Code Extension

Empower Pro Dev users, such as data engineers and data scientists, to securely run custom code extensions from ingest to act across the Data 360's structured and unstructured pipeline

Code Extension Script

Code procedure to customize specific Data 360 feature like Batch Transform

Use secure Data 360 SDK to read or write to Data 360 Objects - DLO, UDLO etc.

Invoked from feature UI like Data Transform & Streaming Transform

Can call endpoint to LLM gateway, Einstein Studio Model & registered External APIs

Code Extension Function

Code function to customize part of the feature at specific extension points

Data passed at execution time using arguments from the calling feature

Invoked from specific extension point within the Feature like Chunking

Can call endpoint to LLM gateway, Einstein Studio Model & registered External APIs

Sample Use Cases

salesforce



Encryption/ Decryption of Sensitive Data

Reduces risks of
moving data to
external platforms



Complex XML Parsing

Provides clean,
structured inputs for
AI agentic workflows



Employee-Role Matching

Creates a more
complex and accurate
matching of resources
with initiatives



Chunking complex documents

Ensures agents
retrieve the most
relevant answers
ensuring AI precision

Development Environment

salesforce

Vibe Code from any Agent

Public SDK (salesforce-data-custom code)

- Read/write DLO, DMO, docs, chunks
- LLM Gateway and Einstein Studio integration
- External API callout wrappers (coming soon)

Install from GitHub or PyPI

DevOps through SF CLI

- `init`, `scan`, `run`, `deploy` commands
- Datakits

Install from npm or download

Code Extensions Skill (sf-skills)

- Full development lifecycle explained
- Best practices & troubleshooting

Install from GitHub

Data 360 MCP Server

- Data 360 Connect REST APIs (transforms, search indexes, monitor...)

Activate in org and connect from agent

Pre-configured in Agentforce Vibes

Development Lifecycle

salesforce

LOCAL ENVIRONMENT

1 Author Code
Using your favorite IDE

2 Run and Debug Locally



3

Deploy code

SANDBOX

4 Run
As batch data transform or
on search index creation

5 Monitor and Debug
DataCustomCodeLogs DLO

6 Package
Using Datakits



7

Install Datakit

PRODUCTION

8 Run
As batch data transform or
on search index creation

9 Monitor
DataCustomCodeLogs DLO

1 Author in your Local Environment

1. Scaffold with init command:

```
sf data-code-extension script init --package-dir ./data-cleaning
```

```
sf data-code-extension function init --package-dir ./text-chunking
```

2. Write transform / chunking logic in `entrypoint.py`

- Call LLMs, predictive models or external APIs through the SDK

3. Run `scan` command to update permissions (`config.json`) and dependencies (`requirements.txt`)

```
sf data-code-extension script scan --entrypoint  
./data-cleaning/payload/entrypoint.py
```

```
sf data-code-extension function scan --entry-point  
./text-chunking/payload/entrypoint.py
```

Batch Transform Script



Entrypoint.py file structure

```
from datacustomcode.client import Client
from datacustomcode.io.writer.base import WriteMode
from pyspark.sql.functions import upper

def main():
    client = Client()
    # IN:  a DMO/DLO
    df = client.read_dlo("Employee__d11")
    df = df.withColumn("position__c", upper(df.position__c))
    # OUT:  a DMO/DLO
    client.write_to_dlo("Employee_Upper__d11", df, write_mode=WriteMode.APPEND)
```

Chunking Function

Entrypoint.py file structure



```
from datacustomcode.function import Runtime
from datacustomcode.function.feature_types.chunking import (
    ChunkType, SearchIndexChunkingV1Request,
    SearchIndexChunkingV1Response, SearchIndexChunkingV1Output,
)

CHUNK_SIZE, OVERLAP = 200, 40

def function(request: SearchIndexChunkingV1Request, runtime: Runtime) -> SearchIndexChunkingV1Response:
    chunks, seq_no = [], 1
    for doc in request.input:                # IN: parsed docs
        words = doc.text.split()
        start = 0
        while start < len(words):
            text = " ".join(words[start:start + CHUNK_SIZE])
            chunks.append(SearchIndexChunkingV1Output(
                text=text, seq_no=seq_no, chunk_type=ChunkType.TEXT))
            seq_no += 1
            start += CHUNK_SIZE - OVERLAP
    return SearchIndexChunkingV1Response(output=chunks)    # OUT: chunks
```

2 Run and Debug in your Local Environment

1. Run using the `run` command:

- a. Transform scripts locally against a data subset (defaults to 1000 rows, never writes back)

```
sf data-code-extension script run --entry-point  
./data-cleaning/payload/entrypoint.py --target-org <org-alias>
```

- b. Chunking functions locally with test-with a `test.json` file

```
sf data-code-extension function run --entry-point  
./text-chunking/payload/entrypoint.py --test-with  
./text-chunking/payload/tests/test.json --target-org <org-alias>
```

3 Deploy

1. Prerequisites:

- Activate (Data 360 setup → Feature Activation → Code Extension)
- Check Docker is running

2. Deploy through the `deploy` command or the UI

```
sf data-code-extension script deploy --name <code-extension-name> --description  
<code-extension-desc> --package-version <package-version> --package-dir  
./data-cleaning/payload --target-org <org-alias> --cpu-size <cpu-size>
```

```
sf data-code-extension function deploy --name <code-extension-name> --description  
<code-extension-desc> --package-version <package-version> --package-dir  
./text-chunking/payload --target-org <org-alias> --cpu-size <cpu-size>
```

4 Run in the Org

- Batch Transform Script:
 1. Create associated Batch transform (auto created in CLI deployments)
 2. Run it
- Chunking Function:
 1. Create a Search Index
 2. Pick Code Extension as the chunking strategy
 3. Build the index
- The Data 360 MCP Server can run these for you and check whether they finished

5

Monitor

1. Anything printed to `stdout` is captured in `DataCustomCodeLogs__dll`. Additionally holds SDK warnings, runtime errors, and missing config alerts

```
SELECT DataCustomCodeName__c, Message__c, ProcessDefinitionName__c,  
Timestamp__c FROM DataCustomCodeLogs__dll
```

2. Query it with the SF CLI, the Query Editor, Data Explorer, or the Data 360 MCP Server

Today's Demo - XML Extraction

Everything headless!

salesforce



Script Transform
Parse XML



Code Extension
Skill



Data 360 MCP
Server



Claude Code

Code Extension - Demo



TerminalShellEditViewWindowHelp

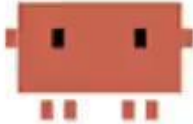
Recently Viewed | Data Stream | Data Cloud Setup | Salesforce | Ask Gemini

claude — Claude Code — mcp-adaptor — claude — 120x36

rohit.dar@rohitda-ltmbpug claude % claude

Claude Code v2.1.236

Welcome back!



Opus Plan · Amazon Bedrock
~/claude

Tips for getting started

Run /init to create a CLAUDE.md file with instructions for Claude

What's new

Added 'maxEffortLevel' setting (top-level or per model under 'modelSettings'): cap...
Added '--system-prompt-snapshot off' to render the system prompt fresh on every re...
Fixed Cowork scheduled tasks in the cloud failing at startup for organizations who...
[/release-notes for more](#)

> █

 Sonnet 5 ● high |  0% | 💰 \$300/\$50000 | 📁 claude | 🌿 no branch

⏏ manual mode on · ← for agents

Initialize Project

Code Extension - Roadmap

GA Now

Q4 2026

Q1 2027

Beyond

- Script Batch Transform
- Function Chunking
- Generative Model (LLM Gateway)
- Predictive Model
- Author to Deploy & promote experience (UI/CLI/Agent)

- Streaming Transform
- External API Calls
- Incremental Transform

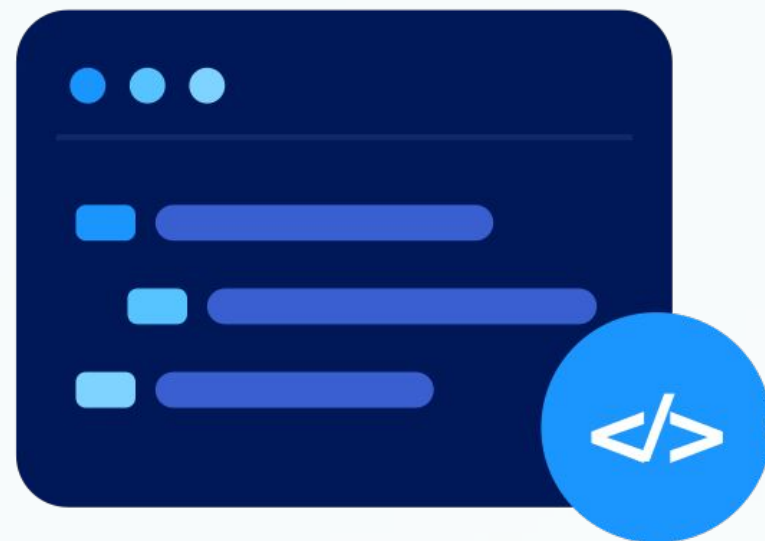
- Code Interpreter
- Query Function
- Run time Parameter
- Unstructured Harmonization

- Script RAG
- Ingestion API
- ISV Support
- Interactive Notebook

Takeaways

salesforce

Code Extensions extend
Data 360 with Python



Tools to handle the full
development lifecycle
with governance built in



Fully headless and GA



Get started today!

Resources





Thank you



Coffee on us.

The first 4,000 attendees to provide feedback on this event will receive a \$5 Starbucks gift card.*

1. Download the Salesforce Events mobile app.
2. Navigate to surveys.
3. Complete (4) session surveys and the Event Survey.
4. Redeem your gift at Badge Pickup in-person on the last day of the event.

*Restrictions apply. See terms and conditions at sforce.co/survey-terms



Session summaries available soon

A short recap of this session is on its way. You'll find it later today in the [Salesforce Events mobile app](#) or online on the [sessions page](#).

